

2 (a) Define acceleration.

(b)

rate of change of velocity	B1 Allow in symbols if symbols defined Allow 'change in velocity per (unit) time' Allow 'change in velocity / time (taken)' Allow 'change in velocity over time' (BOD) Allow 'ratio of change in velocity and time' Not 'rate of change of velocity per unit time' as implies a rate of change of a rate of change. Not 'velocity / time' or 'change of rate of velocity' Not 'change of velocity in unit time' (as no ratio) Not 'change in velocity per second' Ignore 'velocity into time' / 'time into velocity' Ignore 'change in velocity by time' Ignore 'speed' for velocity Ignore any units	 [1]
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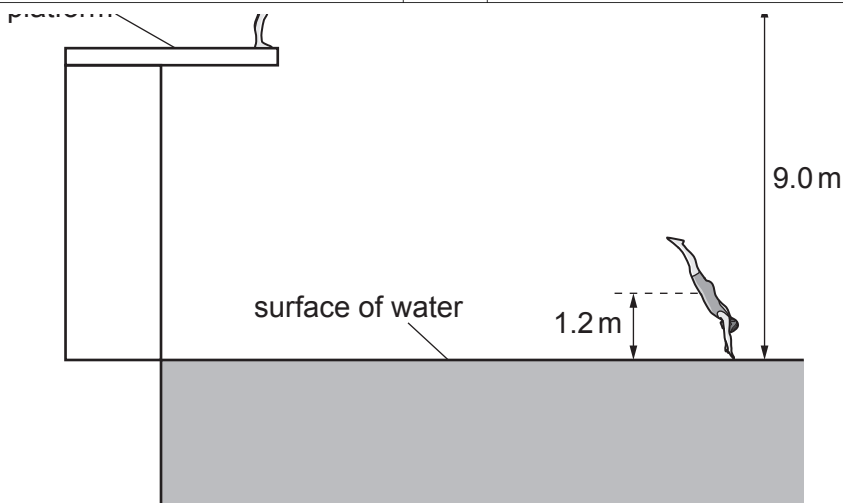


Fig. 2.1 (not to scale)

When the diver is on the platform his centre of gravity is a vertical height of 9.0 m above the surface of the water. The diver jumps from the platform with a velocity of 5.9 m s^{-1} at an angle of 60° to the horizontal.

Air resistance is negligible.

When the diver hits the surface of the water, his centre of gravity is a vertical height of 1.2 m above the surface of the water.

Calculate the speed of the diver at the instant he hits the surface of the water.

$\frac{1}{2} m(\Delta)v^2 = mg(\Delta)h$ $v^2 = 5.9^2 + 2 \times 9.81 \times 7.8$ $v^2 = 188$ $v = 14 \text{ m s}^{-1}$ OR by resolving components Vertically: $v^2 = u^2 + 2as$ $v^2 = (5.9 \sin 60)^2 + 2 \times -9.81 \times (1.2 - 9.0)$ $v_v = 13.4$ horizontally: $v_h = 5.9 \cos 60$ $v_h = 2.95$ resultant velocity = $\sqrt{(13.4^2 + 2.95^2)}$ $= 14 \text{ m s}^{-1}$	C1 Any subject. C1 Allow $g = 10$ as final answer is still 14 m s^{-1} to 2SF. Allow opposite sign convention if consistent. (i.e. $v^2 = 5.9^2 - 2 \times 9.81 \times (1.2 - 9.0)$) If Δh given as 9.0 m treat as XP, so only 1st C1 can be scored A1 Allow $\pm 14 \text{ m s}^{-1}$ as question asks for speed. C1 marks in resolution method are independent of one another. (C1) Allow opposite sign convention if consistent. (i.e. $v^2 = (-5.9 \sin 60)^2 + 2 \times 9.81 \times (9.0 - 1.2)$) Allow C1 even if 13.4 on answer line (working must be seen) (C1) Allow 'v' / 'v_y' for 'v_v'. Allow 'v_x' / 'u_x' / 'u_h' for 'v_h' (A1) alternative calculation for vertical component: s to max height = $[0^2 - (5.9 \sin 60)^2] / (2 \times -9.81) = 1.33$ vertical s from max height to water = $1.2 - 9.0 - 1.33 = (-)9.13$ $v^2 = (0)^2 + 2 \times 9.81 \times 9.13$ (C1) $v_v = 13.4$ If trig functions in both components are wrong way around, treat as AE in the first component and Allow ECF for the second component.	 [3]
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(c) The diver in (b) enters the water and decelerates.

(i) Describe and explain the variation of the viscous drag force acting on the diver in the water as he moves downwards.

(As the diver moves down their) speed decreases	B1	Allow 'velocity' for 'speed'	
(So) viscous force / drag (force) decreases	B1	Allow 'slows down' for 'speed decreases'	
		Allow candidates to describe a decrease in either speed or force in terms of two values e.g. speed goes from 10m/s to 5m/s (as this is clearly a decrease)	
		Allow 'resistive force'/'water resistance' for viscous force	
		Ignore acceleration decreases	
		Ignore any reference to terminal velocity	
		Ignore references to upthrust	[2]

(ii) The diver has a volume of $7.5 \times 10^{-2} \text{ m}^3$. The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$.

Show that the upthrust acting on the diver when he is entirely underwater is 740 N.

$(F =) \rho g V$ $= 1000 \times 9.81 \times 7.5 \times 10^{-2} = 740(\text{N})$	A1	no mark for copying symbol formula from Formulae sheet Not $g = 10$ as gives 750N to 2SF. It is a 'show that' question, so for the A1 mark need to see both the <u>numerical</u> substitution and the final answer to 2 or more SF. If answer given to more than 2SF, it must equal 740 when rounded by the examiner to 2SF. Working can be shown in two parts (both must be seen): $m = 1000 \times 7.5 \times 10^{-2} = 75$ $F = 75 \times 9.81 = 740 \text{ N}$	[1]
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(iii) At a particular instant when the diver is entirely underwater his horizontal velocity is zero. The viscous drag force acting on him at this instant is 950 N vertically upwards. The diver has mass 78 kg.

Determine the magnitude and direction of the acceleration of the diver.

resultant Force = $740 + 950 - (78 \times 9.81)$ $= 925$ acceleration = F / m $= 925 / 78$ $= 12 \text{ ms}^{-2}$ (vertically) upwards	C1 Allow $g=10$ ($F=910\text{N}$) as final answer correct to 2SF Allow ± 925 Both C1 marks may be scored independently C1 Allow any force / mass for this mark A1 Allow $\pm 12 \text{ ms}^{-2}$ for the A1 mark B1 direction must be explicitly stated. Allow 'up' BOD Allow 'in direction of drag force' Allow an answer given as an angle if it's stated with a reference line and a direction e.g. '0 from vertically upwards', or '90 clockwise from horizontal left'. Ignore + or - for the direction. Ignore 'opposite to direction of motion'
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acceleration = ms^{-2}

direction

[4]

[Total: 11]

2 (a) Define velocity.

change in displacement / time (taken)	B1
.....	[1]

(b) A student throws a ball over a vertical wall of height h , as shown in Fig. 2.1.

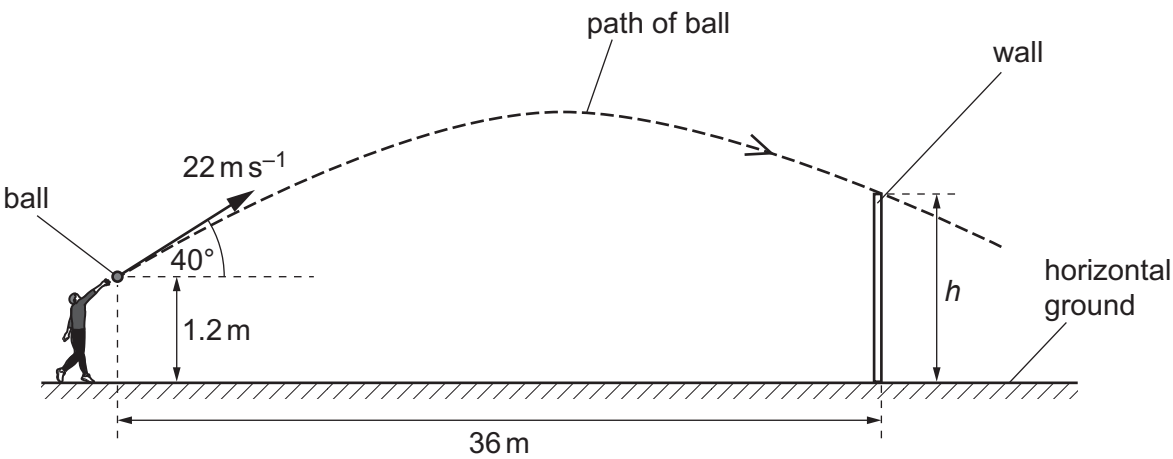


Fig. 2.1 (not to scale)

The ball leaves the hand of the student at a height of 1.2 m above the horizontal ground. The ball has an initial velocity of 22 m s^{-1} at an angle of 40° to the horizontal. The wall is a horizontal distance of 36 m from where the student releases the ball.

Air resistance is negligible.

(i) Determine the time taken for the ball to reach the wall.

horizontal velocity = $22 \times \cos 40^\circ$	C1
time taken = $36 / (22 \times \cos 40^\circ)$ = 2.1 s	A1

time taken = s [2]

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(ii) Calculate the vertical component u of the initial velocity of the ball.

$u = 22 \times \sin 40^\circ$ $= 14 \text{ m s}^{-1}$	A1
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$u = \dots\dots\dots \text{ms}^{-1}$ [1]

(iii) The ball just goes over the wall.

Calculate the height h of the wall.

$s = ut + \frac{1}{2} at^2$ $= (14 \times 2.1) + (\frac{1}{2} \times -9.81 \times 2.1^2)$	C1
$= 7.8 \text{ (m)}$	C1
(therefore) height of wall $= 7.8 + 1.2$ $= 9.0 \text{ m}$	A1

$h = \dots\dots\dots \text{m}$ [3]

[Total: 7]

or	
<i>other methods possible e.g.</i> time to maximum height $= (0 - 14) / -9.81$ ($= 1.43 \text{ s}$) time from maximum height to wall $= 2.1 - 1.43$ ($= 0.67 \text{ s}$) maximum height above release $= (14 \times 1.43) + (\frac{1}{2} \times -9.81 \times 1.43^2)$ $= 9.99 \text{ (m)}$	(C1)
height from maximum to wall $= 0.5 \times 9.81 \times 0.67^2$ ($= 2.2 \text{ m}$) height above release $= 9.99 - 2.2$ $= 7.8 \text{ (m)}$	(C1)
height of wall $= 1.2 + 7.8$ $= 9.0 \text{ m}$	(A1)

- 2 A skydiver jumps from an aircraft at time $t = 0$ and falls vertically downwards. The variation with t of her velocity v is shown in Fig. 2.1.

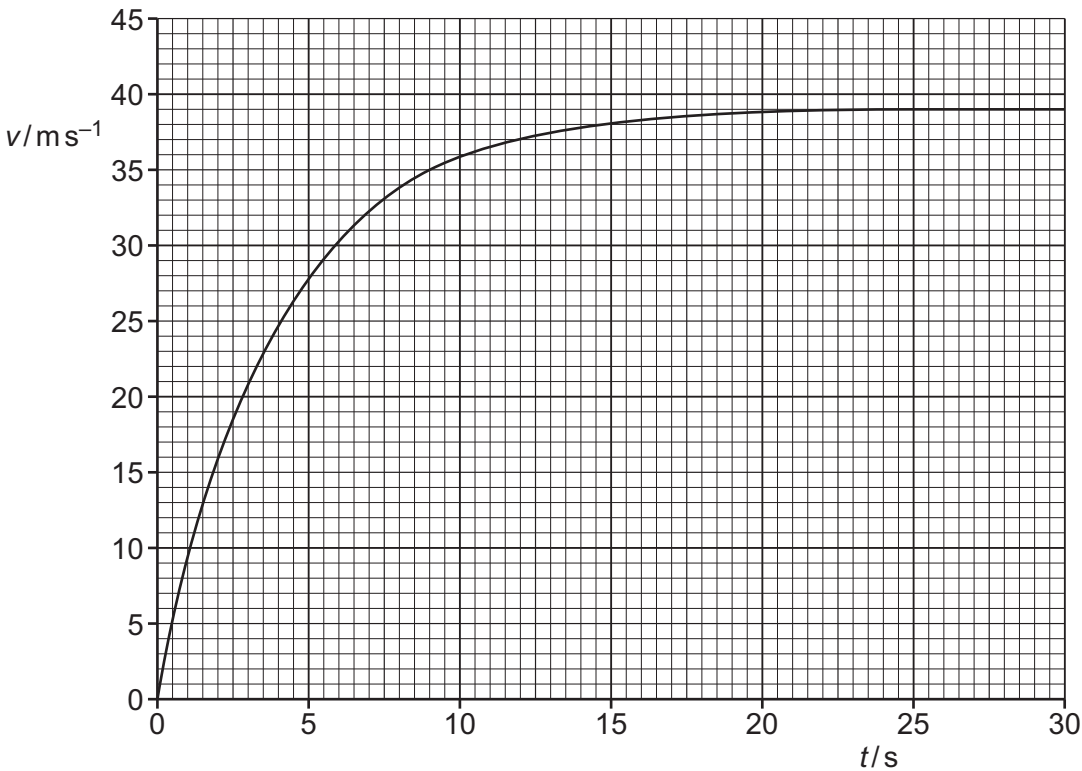


Fig. 2.1

- (a) (i) Using Fig. 2.1, state the terminal velocity of the skydiver.

39 m s ⁻¹	A1	[1]
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- (ii) By drawing a suitable line on Fig. 2.1, determine the acceleration of the skydiver at time $t = 9.0$ s.

tangent line to curve drawn on Fig. 2.1	C1
$a = \text{gradient of tangent line} = \Delta v / \Delta t$ e.g. $= (44 - 26) / (18 - 0)$ $0.9 \leq a \leq 1.1 \text{ m s}^{-2}$	A1

acceleration =m s⁻² [2]

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- (b) The mass of the skydiver and her equipment is 68kg. The upthrust on the skydiver is negligible.

After reaching terminal velocity, the skydiver opens her parachute at time t_1 . A total drag force of 1800 N acts on the skydiver.

Determine the magnitude and direction of the acceleration of the skydiver at time t_1 .

$(\Sigma)F = 68 \times 9.81 - 1800$ $(= -1133 \text{ N})$	C1
$a = (\Sigma)F / m = - 1133 / 68$ $= (-)17 \text{ m s}^{-2}$	A1
upwards	B1

acceleration =ms⁻²

direction =
[3]

- (c) The parachute is fully open at time t_2 . At a later time t_3 the skydiver reaches a constant velocity of 5.7 ms⁻¹.

- (i) Describe and explain the variation with time of the magnitude of her acceleration between time t_2 and time t_3 .

..... drag force decreases (as speed decreases)	B1
(as speed decreases) resultant force decreases so (magnitude of) acceleration decreases (to zero)	B1

.....
.....
..... [2]

- (ii) Calculate the change in momentum of the skydiver between time t_1 and time t_3 .

$(\Delta)p = m\Delta v \text{ or } (\Delta)p = m(v - u)$	C1
$= 68 \times (5.7 - 39)$ $= (-)2300 \text{ N s}$	A1

change in momentum = Ns [2]

[Total: 10]

2 (a) Define displacement from a point.

distance (from the point) in a straight line in a given direction	B1	
..... [1]		

(b) An object is projected horizontally at a speed of 6.0 m s^{-1} from a slope, as shown in Fig. 2.1.

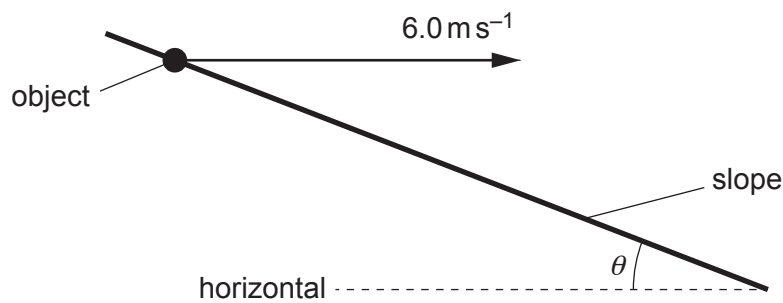


Fig. 2.1 (not to scale)

The slope is at an angle θ to the horizontal. Air resistance is negligible.

The object lands on the slope a time of 0.71 s later and stops without rolling or bouncing.

(i) Determine the horizontal distance travelled by the object.

distance = speed $\times$ time = 6.0×0.71 = 4.3 m	A1
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distance = m [1]

(ii) Determine the vertical distance travelled by the object.

$s = ut + \frac{1}{2}at^2$ = $\frac{1}{2} \times 9.81 \times 0.71^2$	C1
= 2.5 m	A1

distance = m [2]

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(iii) Use your answers in **(b)(i)** and **(b)(ii)** to calculate θ .

$\tan \theta = 2.5 / 4.3$ or hypotenuse = $\sqrt{(4.3^2 + 2.5^2)}$ (= 4.97 m) $\cos \theta = 4.3 / 4.97$ or $\sin \theta = 2.5 / 4.97$	C1
$\theta = 30^\circ$	A1 [2]

(iv) Determine the magnitude of the displacement of the object from its original position.

displacement = $\sqrt{(4.3^2 + 2.5^2)}$	C1
= 4.9 m or 5.0 m	A1
or	
displacement = $2.5 / \sin 30^\circ$ or displacement = $4.3 / \cos 30^\circ$	(C1)
= 5.0 m	(A1)

displacement = m [2]

(v) By considering energy, calculate the speed of the object just before it lands.

KE = $\frac{1}{2}mv^2$ or GPE = mgh	C1
initial KE + loss in GPE = final KE $(\frac{1}{2} \times m \times 6.0^2) + (m \times 9.81 \times 2.5) = (\frac{1}{2} \times m \times v^2)$	C1
$v = 9.2 \text{ m s}^{-1}$	A1

speed = ms^{-1} [3]

[Total: 11]

1 (a) Define acceleration.

rate of change of velocity	B1	
..... [1]		

(b) A small aircraft is flying horizontally at a speed of 42 ms^{-1} at a height of 63 m above horizontal ground, as shown in Fig. 1.1.

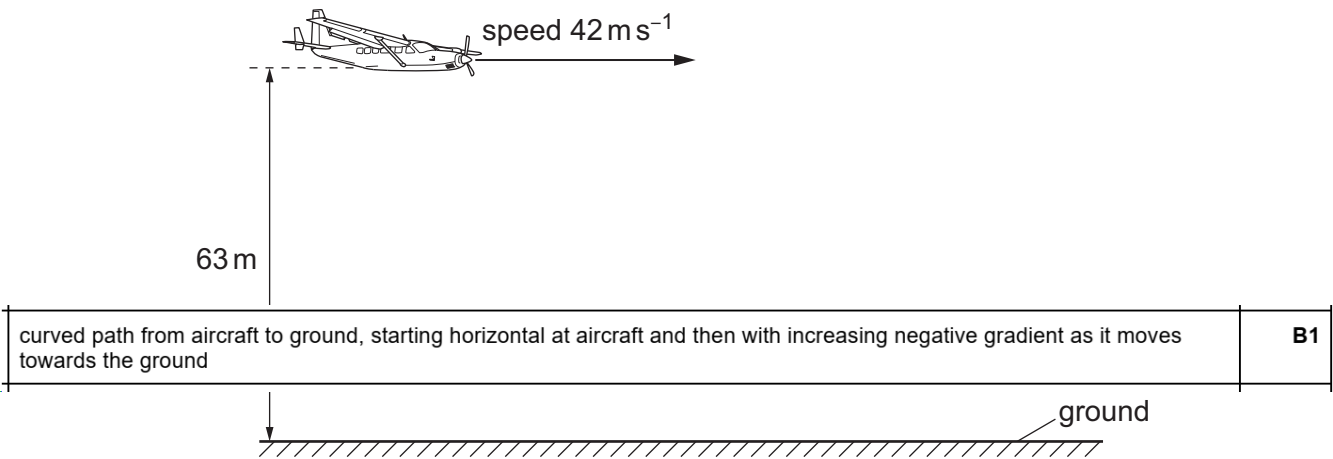


Fig. 1.1

The aircraft drops a small parcel. The parcel is released from the aircraft at the instant shown in Fig. 1.1. Air resistance is negligible.

- (i) On Fig. 1.1, draw a line to show the path of the parcel as it falls from the aircraft to the ground. [1]
- (ii) Calculate the time taken from the instant of release to the instant the parcel reaches the ground.

$s = ut + \frac{1}{2}at^2$	C1
$63 = \frac{1}{2} \times 9.81 \times t^2$	
time = 3.6 s	A1

time = s [2]

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(iii) Calculate the vertical component of the velocity of the parcel immediately before it reaches the ground.

$v^2 = 2 \times 9.81 \times 63$ or $v = 0 + (9.81 \times 3.6)$ or $63 = (v \times 3.6) - (\frac{1}{2} \times 9.81 \times 3.6^2)$ or $63 = \frac{1}{2} \times (0 + v) \times 3.6$ $v = 35 \text{ m s}^{-1}$	A1
vertical component of velocity = ms^{-1} [1]	

(iv) Determine the speed at which the parcel reaches the ground.

$\text{speed}^2 = 35^2 + 42^2$	C1
$\text{speed} = 55 \text{ m s}^{-1}$	A1

speed = ms^{-1} [2]

[Total: 7]